

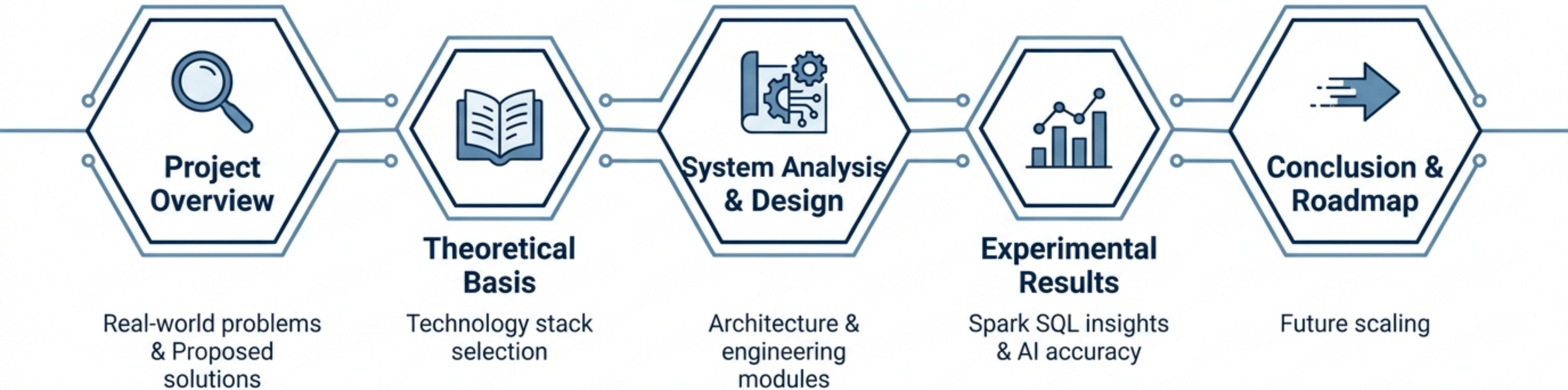
Advanced Machine Learning for Cardiovascular Disease Early Warning Systems

Architecting an End-to-End Big Data Pipeline from Clinical History

VNU-HCM University of Information Technology (UIT)



Mapping the Journey from Raw Clinical Data to Medical Intelligence



The Objective is Automated Triage at Population Scale

Core Objective: Build an end-to-end automated Big Data pipeline optimized for the early warning and triage of at-risk patients.

Anatomy of a Dataset

70,000
Raw Clinical Records



13 Features

1 Target Label



Age



Gender



BMI



Blood Pressure



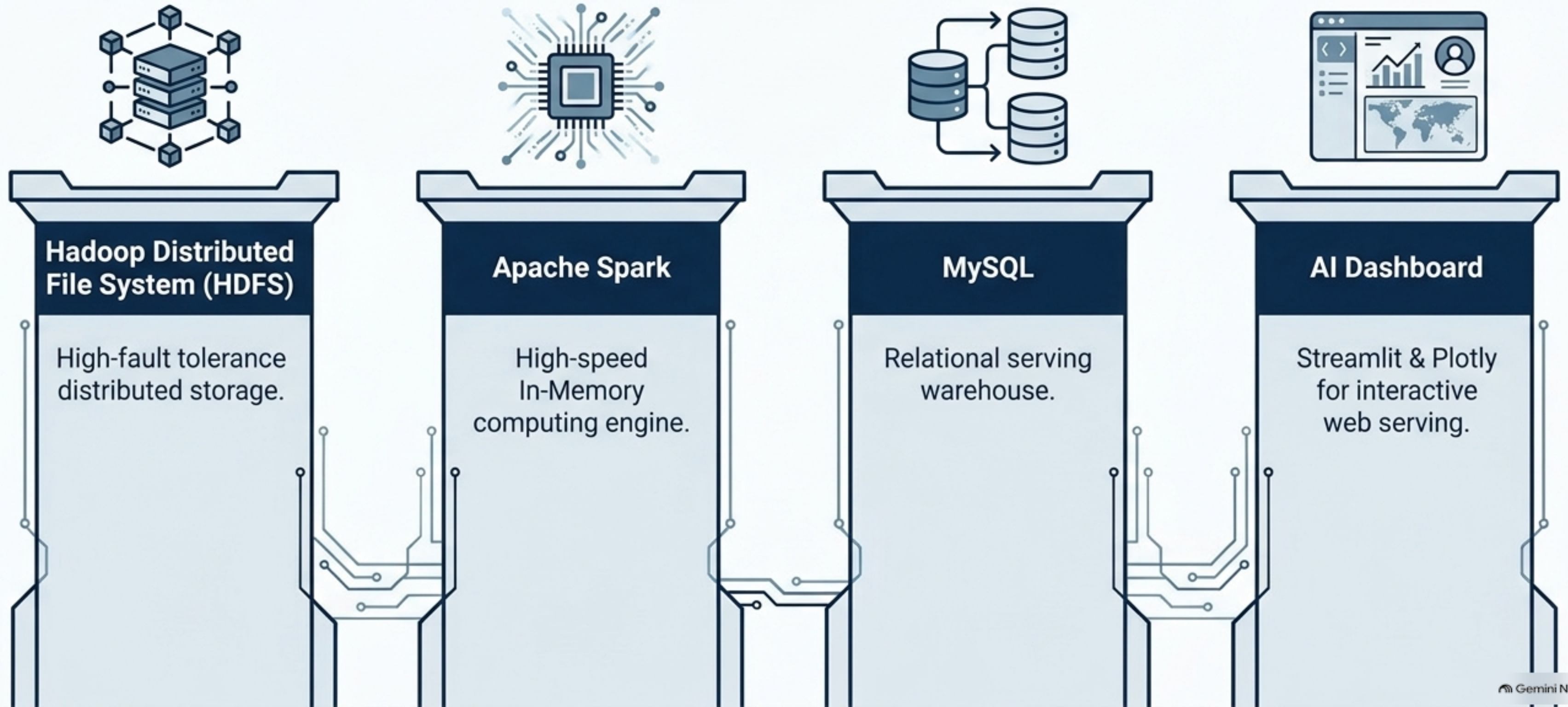
Glucose

Demographics & vitals



0 (Healthy) or 1 (Cardiovascular Disease **Detected**)

The Big Data Ecosystem Powering the Pipeline



Layering Storage, Compute, Serving, and Interface

Storage Core

HDFS & Parquet

Utilizing columnar compression to reduce physical footprint and vastly optimize read/write I/O operations.



Compute Engine

Apache Spark

Leveraging In-Memory execution for rapid ETL processing and PySpark MLlib for distributed machine learning.



Serving Layer

MySQL Server

Acting as a lightweight Data housing 1 master table and 9 aggregated epidemiological risk tables.



User Interface

Streamlit + Plotly

An interactive web framework visualizing medical data and serving as a real-time Doctor AI interface.



Translating Computing Power into Clinical Value



High-Speed Processing

In-Memory execution allows the rapid parsing and analysis of 70,000 clinical records simultaneously.



Seamless Automation

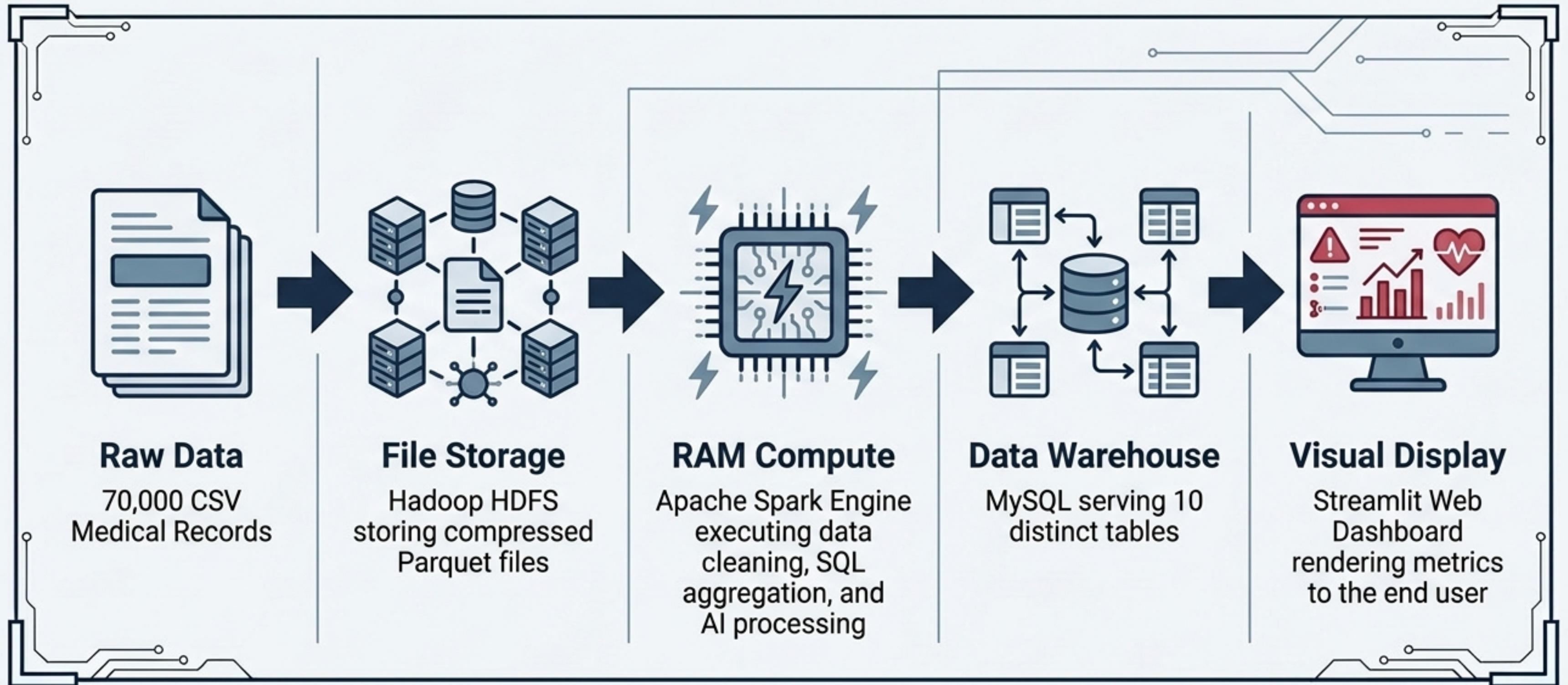
Complete automation of the data flow from raw CSV cleaning, through risk aggregation, to Data Warehouse insertion.



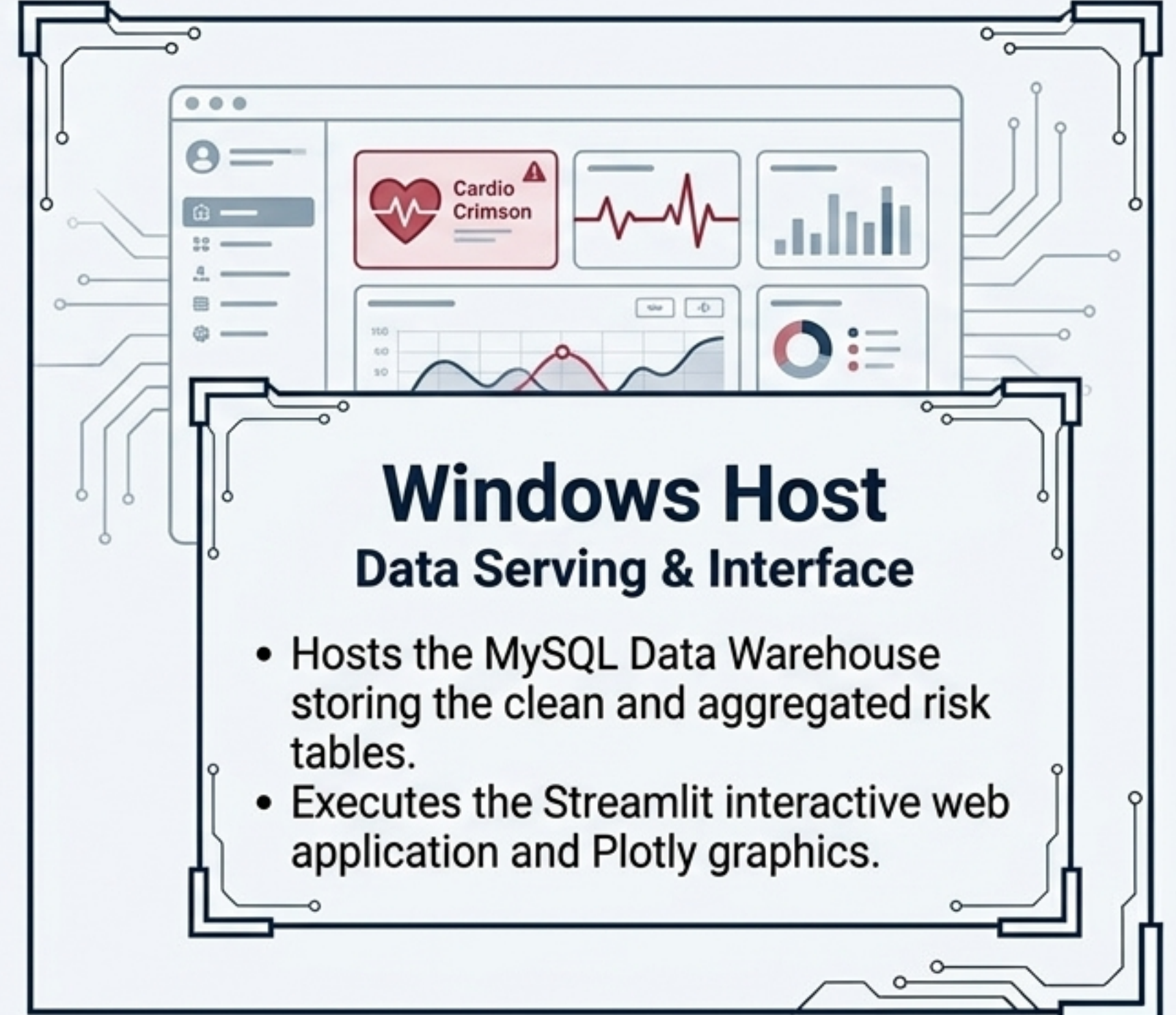
Clinical Decision Support

Delivers instant AI diagnosis, identifying high-risk community groups, and generating actionable triage recommendations.

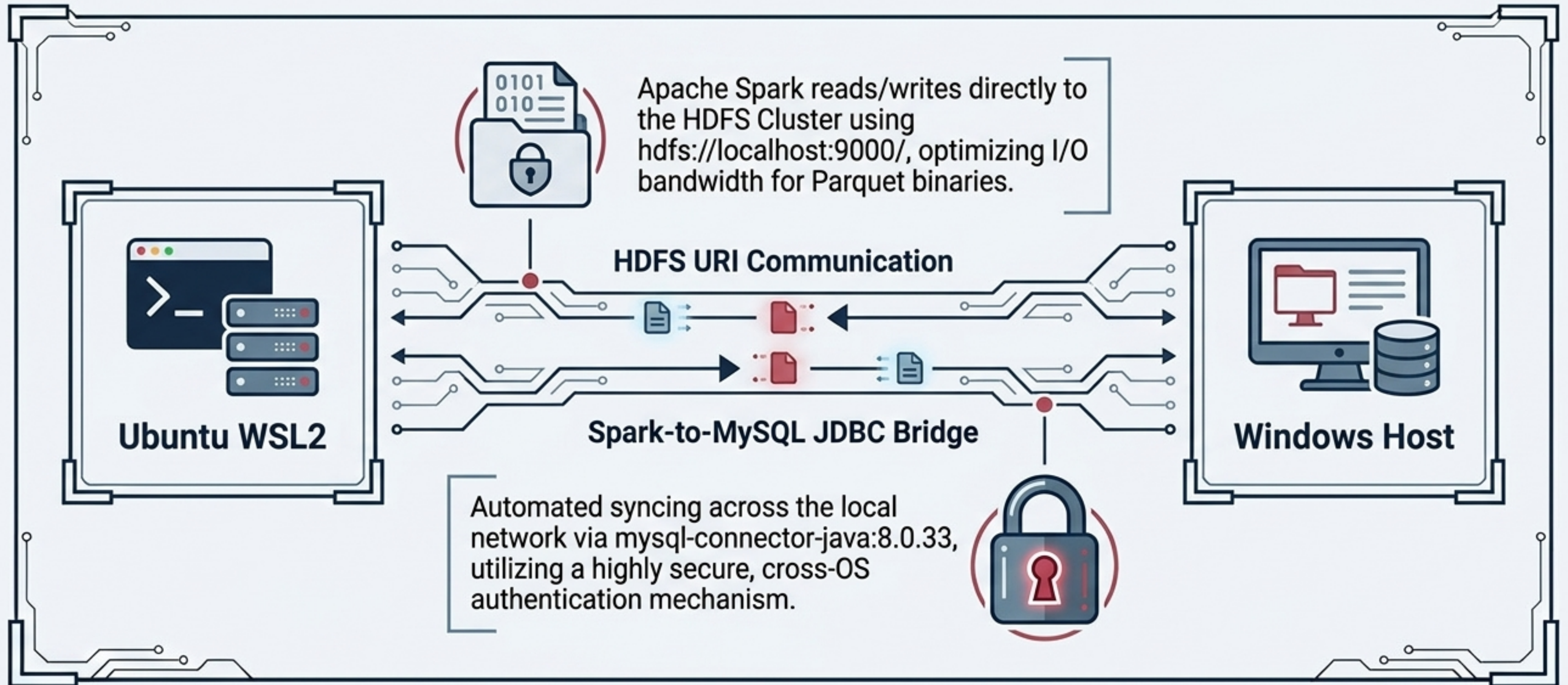
System Architecture and Data Flow Diagram



Multi-Platform Deployment Environment



Cross-OS Communication Protocols and Setup



The Software Engine:

Data Cleaning and Analysis Modules

MODULE 01

`clean_data.py`

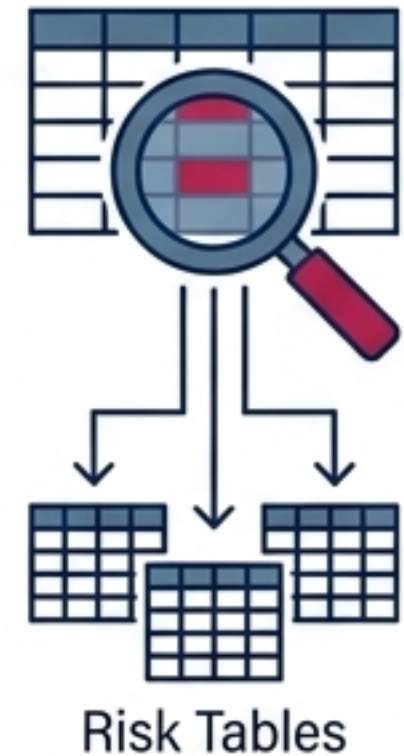
Ingests raw CSV data.
Filters out biological noise
and impossible vital signs.
Generates new `age_years`
and `BMI` features.
Exports highly
compressed Parquet files.



MODULE 02

`analyze_data.py`

Leverages Spark SQL
to query the clean
Parquet data,
aggregating 13 clinical
features into 9
independent
epidemiological risk
tables.

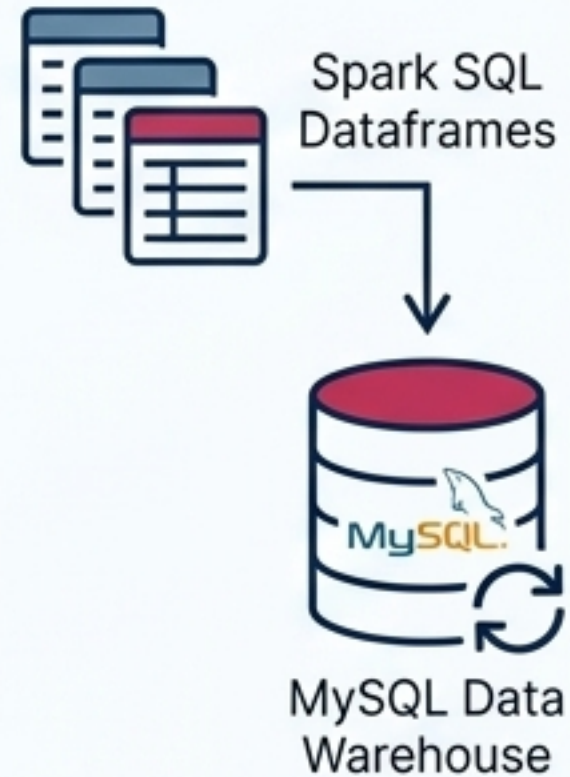


The Software Engine: Warehouse Sync and Distributed AI

MODULE 03

`export_mysql.py`

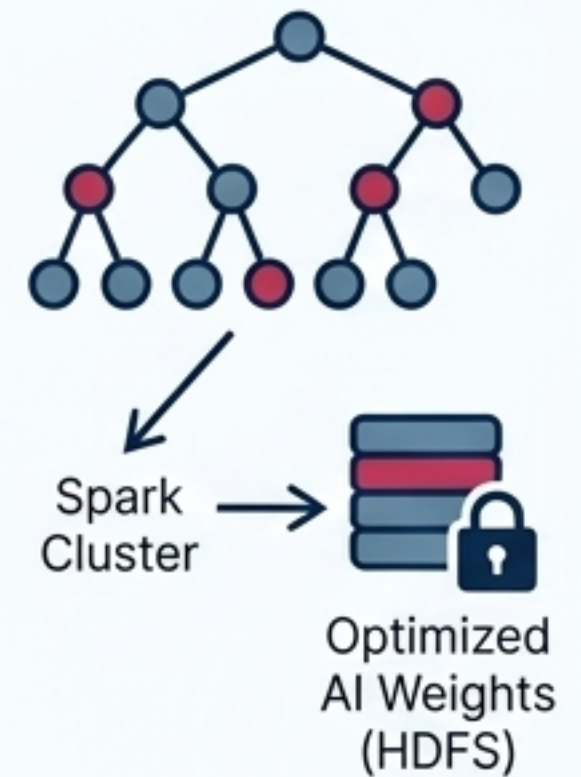
Automatically pushes the aggregated Spark SQL dataframes into the MySQL Data Warehouse via the JDBC bridge.



MODULE 04

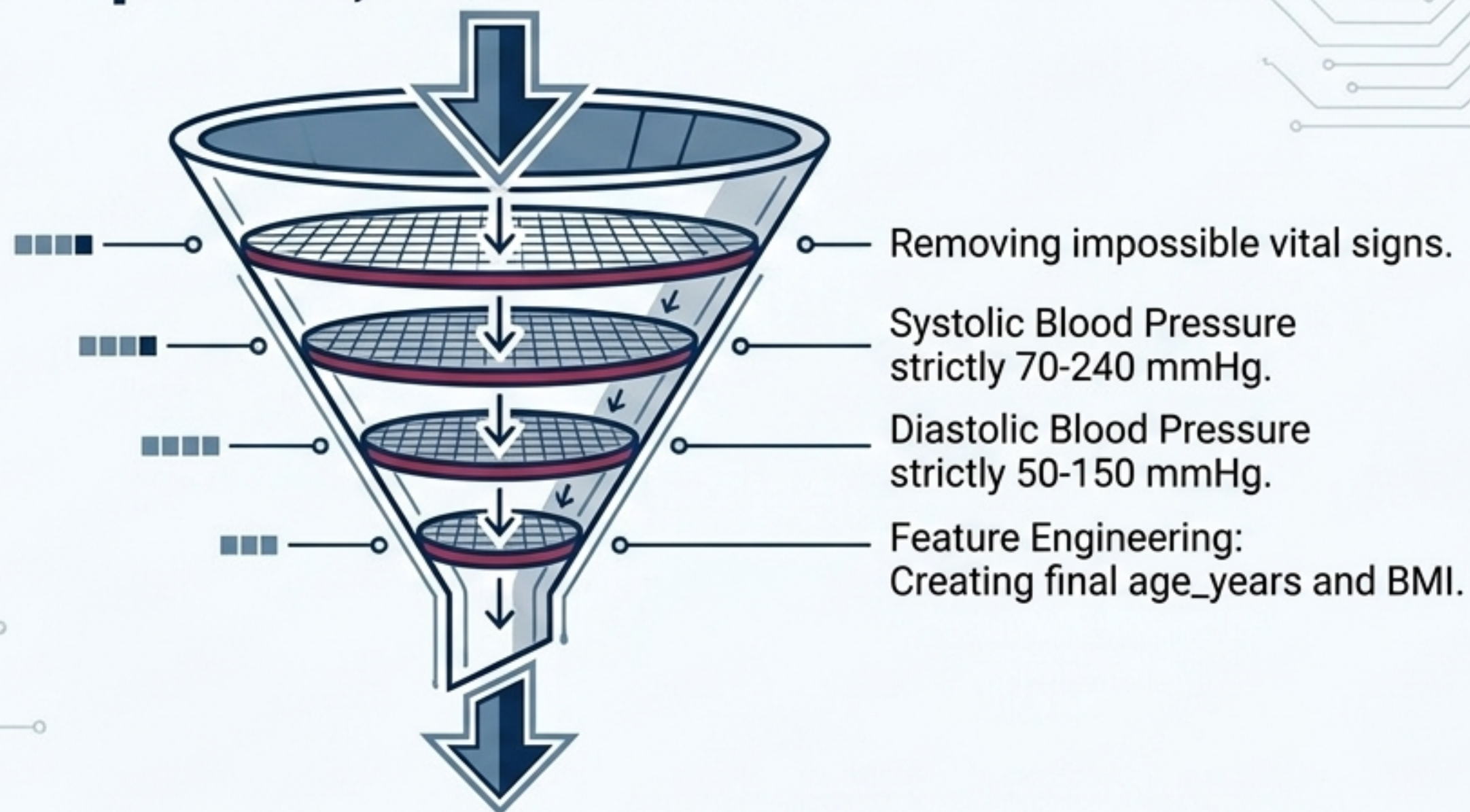
`train_ai.py`

Initiates distributed training of the Random Forest machine learning model on the Spark cluster, ultimately saving the optimized AI weights back to HDFS.



Imposing Strict Biological Filtering Criteria

Input: 70,000 Raw Records



Output: 68,614 Clean, standard-compliant clinical records ready for machine learning.

Experimental Results: Spark SQL Risk Differentials

	RISK FACTOR & DIFFERENTIAL ANALYSIS	RISK MULTIPLIER
Blood Pressure	Baseline 22.15% (Normal) vs. 80.14% (Stage 2 HTN). +57.99% differential.	~3.6x Risk Multiplier
Age	Baseline 21.55% (<40yo) vs. 68.86% (>60yo). +47.31% differential.	~3.2x Risk Multiplier
Cholesterol	Baseline 43.54% (Normal) vs. 76.26% (High). +32.72% differential.	~1.75x Risk Multiplier
BMI	Baseline 39.73% (Normal) vs. 62.46% (Obese). +22.73% differential.	~1.57x Risk Multiplier

Minor Factors: Glucose showed a +14.21% increase; Smoking showed a negligible -2.86% differential.

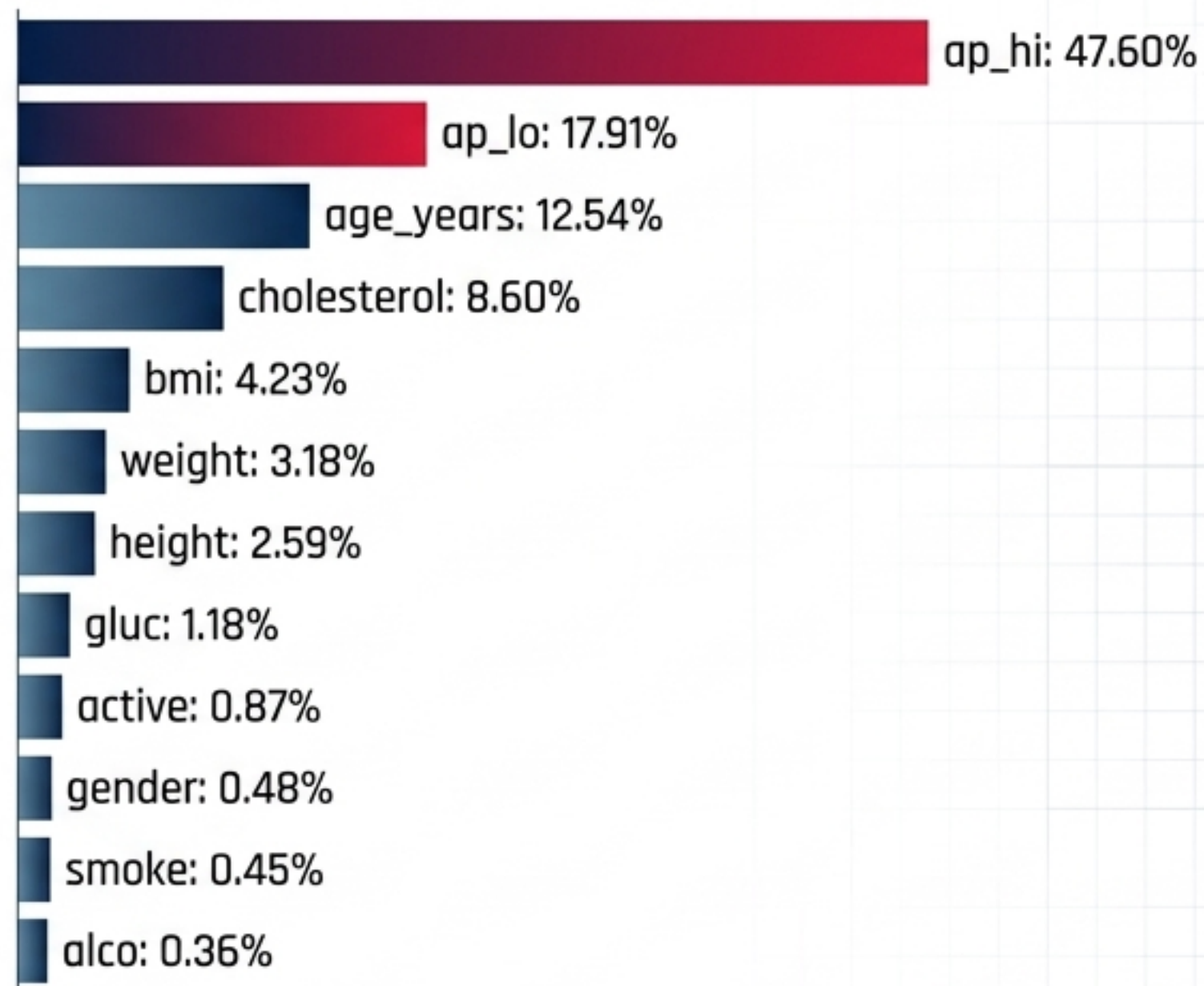
AI Model Evaluation and Feature Importance

Performance Metrics



AUC-ROC = 79.94%

Feature Weights Breakdown



Insight: Blood Pressure (ap_hi + ap_lo) commands >65.5% of the AI's decision-making power, perfectly corroborating the Spark SQL risk differentials.

Synthesis: The Four Core Drivers of Cardiovascular Risk



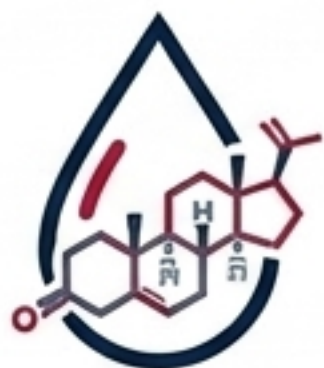
Blood Pressure

The dominant predictor, **>65% AI weight**



Age

The steepest demographic risk curve



Cholesterol

The leading biochemical marker



BMI

The primary lifestyle-modifiable factor

Interface Deployment: Statistical Dashboard (Tab 1)

Function: A visual administration panel querying the MySQL Data Warehouse using interactive Plotly charts.



Interface Deployment: Real-Time Doctor AI (Tab 2)

Function: Accepts real-time vital sign inputs from new patients, triggering instant inference via the embedded Spark MLlib model.

The screenshot shows a web browser window titled "Real-Time Doctor AI (Tab 2)" with the URL "internal.network.12.0.1.072006/noctes/forms/". The form contains the following fields:

- Tuổi: 48
- Giới tính: ☒ Nam ☐ Nữ
- Chiều cao (cm): 170
- Cân nặng (kg): 80
- Huyết áp tối đa: 150
- Huyết áp tối thiểu: 90
- Cholesterol: ☐ 1: normal ☒ 2: above normal ☐ 3: well above normal
- Glucose: ☒ 1: normal ☐ 2: above normal ☐ 3: well above normal
- Hút thuốc: ☒ Yes ☐ No
- Uống rượu: ☐ Yes ☒ No
- Hoạt động thể chất: ☐ Yes ☒ No

A red bar displays the result: "Xác suất mắc bệnh do AI đánh giá: 85.77%".

Below the bar are three recommendation boxes:

- Chế độ ăn uống:** (with heart icon) Tuân thủ chế độ ăn ít muối, giảm chất béo bão hòa và tăng cường rau xanh, trái cây.
- Kiểm soát cân nặng:** (with scale icon) Duy trì chỉ số khối cơ thể (BMI) ở mức hợp lý thông qua việc cân bằng năng lượng nạp vào và tiêu hao.
- Tập thể dục:** (with running icon) Tập thể dục đều đặn ít nhất 30 phút mỗi ngày, hầu hết các ngày trong tuần, để cải thiện sức khỏe tim mạch.

Expert System Output

- **Visual Alerts:** Generates clear percentage probabilities with distinct Red/Green color coding.
- **Medical Recommendations:** Automatically triggers personalized, actionable advice based on the inputs (low-salt diet, weight control, exercise).

Project Conclusions and Technical Achievements



Core Technology Mastery

Successfully engineered 4 independent Python modules and managed a complex Cross-OS pipeline (WSL2 Ubuntu to Windows).



Architecture Optimization

Fully exploited In-Memory computation (Apache Spark) and high-speed columnar storage (Parquet) to eliminate data bottlenecks.



Practical Application

Transitioned from raw code to a fully deployable Web Application designed for clinical triage and community health screening.

Future Development and Scaling Roadmap

Next Step (Streaming Ingestion)



Integrating Apache Kafka to process 24/7 continuous data streams from IoT smart wearables (heart rate, live BP monitoring).

Scaling Up (Cloud Hadoop Clusters)



Migrating from local deployment to multi-node distributed cloud environments (AWS EMR or GCP Dataproc) to process Terabyte-scale hospital datasets.

Model Refinement



Upgrading the AI engine with larger, standardized datasets to output even more granular medical recommendations.

Foundational References and Literature



- [1] S. Ulianych, 'Cardiovascular Disease Dataset,' Kaggle, 2019.
- [2] M. Zaharia et al., 'Apache Spark: A unified engine for big data processing,' Communications of the ACM, 2016.
- [3] K. Shvachko et al., 'The Hadoop Distributed File System,' MSST, 2010.
- [4] L. Breiman, 'Random forests,' Machine Learning, 2001.
- [5] P. K. Whelton et al., '2017 ACC/AHA Guideline for High Blood Pressure,' JACC, 2018.

Thank You For Your Attention

Q&A Session

